Alexander Alemi

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papers: google scholar
github: alexalemi

Current position

Member of Technical Staff, Anthropic

Previous experience

2018-2025 *Senior Research Scientist*, Google, Machine Perception
2016-2018 *Software Engineer*, Google, Mountain View, CA.
2015-2016 *Postodoc*, Disney Research, Boston
2009-2015 *Research Associate*, Laboratory of Atomic and Solid-State Physics (LASSP), Cornell
2015 *Research Intern*, Google, Mountain View, CA.
2014 *Research Intern*, Disney Reseach, Boston.

Areas of specialization

Information Theory & Deep Learning • Physics • Machine Learning • Python Programming

Education

2015 PhD in Physics, Cornell with Jim Sethna
2013 MSc in Physics, Cornell
2009 BSc in Physics, California Institute of Technology (Caltech)

Publications

2024c K Everett, L Xiao, M Wortsman, AA Alemi, R Novak, PJ Liu, I Gur, *et al.*, "Scaling Exponents Across Parameterizations and Optimizers", arXiv:2407.05872

2024b B Lester, J Lee, A Alemi, J Pennington, A Roberts, J Sohl-Dickstein, *et al.*, "Training LLMs over Neurally Compressed Text", arXiv:2404.03626

2024a D Ho, E Jang, M Khansari, YQ Du, AA Alemi, "Using Embeddings, Generated Using Robot Action Models, in Controlling Robot to Perform Robotic Task", US Patent App. 18/102,053

2023e A Singh, JD Co-Reyes, R Agarwal, A Anand, P Patil, X Garcia, PJ Liu, *et al.*, "Beyond Human Data: Scaling Self-Training for Problem-Solving with Language Models", arXiv:2312.06585

2023d CD Freeman, L Culp, A Parisi, ML Bileschi, GF Elsayed, A Rizkowsky, *et al.*, "Frontier Language Models are not Robust to Adversarial Arithmetic, or 'What do I need to say so you agree 2+2=5?' ", arXiv:2311.07587

2023c M Wortsman, PJ Liu, L Xiao, K Everett, A Alemi, B Adlam, JD Co-Reyes, *et al.*, "Small-Scale Proxies for Large-Scale Transformer Training Instabilities", arXiv:2309.14322

2023b I Seroussi, AA Alemi, M Helias, Z Ringel, "Speed Limits for Deep Learning", arXiv:2307.14653

2023a AA Alemi, B Poole, "Variational Prediction", arXiv:2307.07568

2022c WR Morningstar, AA Alemi, JV Dillon, "PAC^m-Bayes: Narrowing the Empirical Risk Gap in the Misspecified Bayesian Regime", *AISTATS 2022* arXiv:2110.09629

2022b Y Ruan, S Singh, W Morningstar, AA Alemi, S Ioffe, I Fischer, JV Dillon, "Weighted Ensemble Self-Supervised Learning", arXiv:2211.09981

2022a Y Du, D Ho, A Alemi, E Jang, M Khansari, "Bayesian Imitation Learning for End-to-End Mobile Manipulation", *ICML 2022*

2021b S Stanton, P Izmailov, P Kirichenko, AA Alemi, AG Wilson, "Does Knowledge Distillation Really Work?", *NeurIPS 2021*

2021a I Korshunova, D Stutz, AA Alemi, O Wiles, S Gowal, "A Closer Look at the Adversarial Robustness of Information Bottleneck Models", arXiv:2107.05712

2020c WR Morningstar, C Ham, AG Gallagher, B Lakshminarayanan, AA Alemi, JV Dillon "Density of States Estimateion for Out-of-Distribution Detection", arXiv:2006.09273

2020b DS Karls, M Bierbaum, AA Alemi, RS Elliot, JP Sethna, EB Tadmor "The OpenKIM Processing Pipeline: A Cloud-Based Automatic Materials Property Computation Engine", *ChemPhys* arXiv:2002.05380

2020a I Fischer, AA Alemi "CEB Improves Model Robustness", arXiv:2002.05380

2019h R Novak, L Xiao, J Hron, J Lee, AA Alemi, J Sohl-Dickstein, SS Schoenholz "Neural Tangents: Fast and Easy Infinite Neural Networks in Python", *ICLR 2020* arXiv:1912.02803

2019g AA Alemi "Variational Predictive Information Bottleneck", arXiv:1910.10831

2019f R Shwartz-Ziv, AA Alemi "Information in Infinite Ensembles of Infinitely-Wide Networks", *AABI* arXiv:1911.09189

2019e Z Dong, D Oktay, B Poole, AA Alemi "On Predictive Information in RNNs", arXiv:1910.09578

2019e T Conte, E DeBenedictis, N Ganesh, T Hylton, JP Strachan, RS Williams, AA Alemi, L Altenberg, G Crooks, J Crutchfield, L del Rio, J Deutsch, M DeWeese, K Douglas, M Esposito, M Frank, R Fry, P Harsha, M Hill, C Kello, J Krichmar, S Kumar, SC Liu, S Lloyd, M Marsili, I Nemenman, A Nugent, N Packard, D Randall, P Sadowski, N Santhanam, R Shaw, A Stieg, E Stopnitzky, C Teuscher, C Watkins, D Wolpert, J Yang, Y Yufik, "Thermodynamic Computing", *CCC* arXiv:1911.01968

2019d B Seybold, E Fertig, A Alemi, I Fischer, "Dueling Decoders: Regularizing Variational Autoencoder Latent Spaces", arXiv:1905.07478

2019c I Fischer, A Alemi, JV Dillon, TFP Team, "Variational Autoencoders with Tensorflow Probability Layers", *Tensorflow Blog*

2019b B Poole, S Ozair, A van den Oord, AA Alemi, G Tucker, "On Variational Bounds of Mutual Information", *ICML 2019* arXiv:1905.06922

2019a CB Clement, M Bierbaum, KP O'Keeffe, AA Alemi, "On the Use of ArXiv as a Dataset", *ICLR 2019 workshop RLGM* arXiv:1905.00075

2018f AA Alemi, JV Dillon, I Fischer, "Uncertainty in the Variational Information Bottleneck", *UAI 2018 workshop on Uncertainty in Deep Learning, contributed oral* arXiv:1807.00906

2018e AA Alemi, I Fischer, "TherML: Thermodynamics of Machine Learning", *ICML 2018 Workshop, contributed oral.*

2018d AA Alemi, I Fischer, "GILBO: One Metric to Measure Them All", *NIPS 2018 Spotlight* arXiv:1802.04874

2018c JV Dillon, I Langmore, D Tran, E Brevdo, S Vasudevan, D Moore, B Patton, A Alemi, M Hoffman, RA Saurous, "TensorFlow Distributions", *arXiv:1711.10604*

2018b S Abu-El-Haija, B Perozzi, R Al-Rfou, A Alemi, "Watch your step: Learning graph embeddings through attention", *NIPS 2018 Poster* arXiv:1710.09599

2018a AA Alemi, B Poole, I Fischer, JV Dillon, RA Saurous, K Murphy, "Fixing a Broken ELBO", *Talk ICML 2018* arXiv:1711.00464

2017e LX Hayden, AA Alemi, PH Ginparg, JP Sethna, "Jeffrey's prior sampling of deep sigmoidal networks", *arXiv: 1705.10589*

2017d K Fragkiadaki, J Huang, A Alemi, S Vijayanarasimhan, S Ricco, R Sukthankar, "Motion Prediction Under Multimodality with Conditional Stochastic Networks" *arXiv: 1705.02082*

2017c M Bierbaum, BD Leahy, AA Alemi, I Cohen, JP Sethna, "Light Microscopy at Maximal Prediction", *PRX 7 (4), 041007*, arXiv: 1702.07336

2017b C Szegedy, S Ioffe, V Vanhoucke, AA Alemi, "Inception-v4, Inception-ResNet and the Impact of Residual Connections on Learning." *AAAI, 4278-4284*, arXiv: 1602.07261

2017 AA Alemi, I Fischer, JV Dillon, KP Murphy, "Deep Variational Information Bottleneck", *ICLR 2017*, arXiv: 1612.00410

2016c R Shin, AA Alemi, G Irving, O Vinyals, "Tree-Structured Variational Autoencoder" *ICLR 2017 Submission*

2017 B Poole, AA Alemi, J Sohl-Dickstein, A Angelova, "Improved generator objectives for GANS" *NIPS GAN Workshop 2016*, arXiv: 1612.02780

2016 G Irving, C Szegedy, AA Alemi, N EEn, F Chollet, J Urban, "Deepmath-deep sequence modesl for premise selection". *NIPS 2016*, arXiv: 1606.04442

2015g CJM Mathy, F Gonda, D Schmidt, N Derbinsky, AA Alemi, J Bento, FM Delle Fave, JS Yedidia, "SPARTA: Fast global planning of collision-avoiding robot trajectories". Learning, Inference and Control of Multi-Agent Systems Workshop, NIPS 2015

2015f AA Alemi, "Zombies Reading Segmented Graphic Articles on the ArXiv". PhD Thesis, Cornell University

2015e JM Cashore, X Zhao, AA Alemi, Y Liu, PI Frazier, "Clustering via Content-Augmented Stochastic Blockmodels". arXiv: 1505.06538

2015d LX Hayden, R Chachra, AA Alemi, PH Ginsparg, JP Sethna, "Canonical Sectors and Evolution of Firms in the US Stock Markets". arXiv: 1503.06205. Submitted to PLoS one.

2015c AA Alemi, P Ginsparg, "Text Segmentation Based on Semantic Word Embeddings", arXiv: 1503.05543

2015b AA Alemi, M Bierbaum, CR Myers, JP Sethna, "The Statistical Mechanics of Zombies", *Oral Presentation* APS March Meeting 2015

2015a AA Alemi, M Bierbaum, CR Myers, JP Sethna, "You Can Run, You Can Hide: The Epidemiology and Statistical Mechanics of Zombies", arXiv: 1503.01104. Phys Rev E 92. 052801

2014b A Taloni, AA Alemi, E Ciusani, JP Sethna, S Zapperi, CAM La Porta, "Mechanical Properties of Growing Melanocytic Nevi and the Progression to Melanoma", *PloS one*, 9 (4), e94229

2014a A Alemi, R Chachra, P Ginsparg, J Sethna, "Finding Structure in the ArXiv", *Oral Presentations* APS March Meeting 2014

2013b MM Baraldi, AA Alemi, JP Sethna, S Caracciolo, C La Porta, S Zapperi, "Growth and Form of Melanoma Cell Colonies", *Journal of Statistical Mechanics: Theory and Experiment*, 2013, 02, p02032

2013a PY Huang, S Kurasch, JS Alden, A Shekhawat, AA Alemi, PL McEuen, JP Sethna, UTE Kaiser, DA Muller, "Imaging Atomic Rearrangements in Two-Dimensional Silica Glass: Watching Silica's Dance", *Science* 342, 6155 p224-227

2012 A Alemi, J Sethna, "A Group Theoretic Approach to Nonlinear and Gradient Elastic Terms for

Graphene and Carbon Nanotubes", *Oral Presentation, APS March Meeting* 57

2011 RS Ottens, V Quetschke, S Wise, AA Alemi, R Lundoc, G Mueller, DH Reitze, DB Tanner, BF Whiting, "Near-field radiative heat transfer between macroscopic planar surfaces", *Physical Review Letters* 107 (1), 014301

2006 A Alemi, D Stevenson, "Why Venus has No Moon", *Oral Presentation, AAS Meeting* 28, 491

Talks

2026 "Information Theory for Representation Learning", Emory TBIO Journal Club

2025 "EP15: The Information Bottleneck and Scaling Laws with Alex Alemi", Information Bottleneck Podcast

2023g "Information Theory for Representation Learning", InfoCog Workshop @ NeurIPS 2023

2023f "What's Missing? A Speculative Sketch of the Future of Machine Learning and Science", ML and the Physical Sciences Workshop @ NeurIPS 2023

2023e "How to Think About AI", Osceola Neovates

2023d "Variational Prediction", AABI 2023

2023c "A Tale of Two Worlds: The Variational Approach to ML", UCF CRCV

2023b "Introduction to Statistics through Randomization", Google Course

2023a "Inferential Engines", Theoretical Physics for Machine Learning, Aspen

2021e "PAC^m-Bayes", Your Model is Wrong Workshop, NeurIPS 2021

2021d "Order of Magnitude Physics", Google Course

2021c "Machine Learning and Thermodynamics", Scientific Machine Learning Mini-Course (SciML) @ CMU

2021b "VIB is Half Bayes", Advances in Approximate Bayesian Inference Symposium 2021

2021a "Basics of Information Theory", Google Course

2020c "Machine Learning and Thermodynamics", University of Maryland Informal Statistical Physics Seminar

2020b "TherML", American Physical Society Topical Group on Data Science

2020a "Variational Predictive Information Bottleneck", Information Theory and Applications Workshop

2019b "A Case for Compression", NeurIPS 2019 Workshop on Information Theory and ML

2019a "TherML", Aspen: Machine Learning and Physics

2018d "Focusing on the Representation", Cornell AI Seminar

2018c "Thermodynamics and Machine Learning", Cornell Physics Colloquium

2018b "Uncertainty in VIB", UAI UDL Workshop 2018

2018a "Fixing a Broken ELBO", ICML 2018

Community Contributions

2024- Take Stock in Children Mentor

2023-2026 Planning Advisory Board, City of Kissimmee, FL

2022- Action Editor at TMLR

2021 Area Chair NeurIPS

2020 Program Committee, UDL Workshop

Grants, honors & awards

2010 Stephen and Margery Russel Distinguished Teaching Award
2009 Upperclass Merit Award
2008 Upperclass Merit Award
2007 Green Memorial Prize
2005 National Merit Scholar - Siemen's Scholar

Teaching & Outreach

2023 Brainiversity - Introduction to Statistics through Randomization
2022 Brainiversity - Order of Magnitude Physics
2021 Brainiversity - Information Theory
2015 GRASSHOPR - Computing with Bins and Beans
2014 Expanding Your Horizions - The Physics of Bubbles
2013 Expanding Your Horizons - The Physics of Bubbles
2012 Expanding Your Horizons - The Physics of Bubbles
2012 TA - Physics 2218 - Physics III: Thermodynamics, Statistical Mechanics and Wave Phenomenon
2011 Physics Department Teaching Assistant Training Coordinator
2011 Expanding Your Horizons - The Physics of Bubbles
2011 GRASSHOPR - … And Physics for All
2010 Physics Department Teaching Assistant Training Trainer grader - Physics 3317 - Applications of Quantum Mechanics
2010 TA - Physics 2217 - Physics II: Electricity and Magnetism
2009 TA - Physics 2213 - Physics II: Heat/Electromagnitism